Pierre Vandenhove

Premier Assistant at UMONS

Avenue Victor Maistriau 15
B-7000 Mons, Belgium
+32 65 37 35 15
pierre.vandenhove@umons.ac.be
pierre-vandenhove.github.io/
pvdhove

Last updated on June 14, 2026.

Positions

Sept 2024–Present — ***Premier Assistant***, *UMONS – Université de Mons, Belgium*
Belgian faculty position with research and teaching duties. Affiliated with the Theoretical Computer Science Lab in the Computer Science Department.

Oct 2023–Sept 2024 — **Postdoctoral Researcher**, *LaBRI, Université de Bordeaux, France*
Worked with Nathanaël Fijalkow on the ANR project *G4S – Games for Synthesis*.

Education

Oct 2019–Sept 2023 — **PhD in Computer Science**, *UMONS, Belgium; Laboratoire Méthodes Formelles (LMF), Université Paris-Saclay, CNRS, ENS Paris-Saclay, France*
Funded by an F.R.S.-FNRS *Research Fellowship*. Project *FrontieRS: Frontiers of Many-sided Reactive Synthesis*. See "PhD Thesis" section below for more details.

2017–2019 — **Master's degree in Mathematics, specialization in Computer Science**, *UMONS*, Belgium, *Summa cum laude*
Exchange year at *Durham University*, UK in 2017–2018. Master's thesis (2019): *Reachability in Stochastic Hybrid Systems*, supervised by T. Brihaye and M. Randour.

2014–2017 — **Bachelor's degree in Mathematics**, *UMONS*, Belgium, *Summa cum laude*
Academic minor in Computer Science.

Grants

2026–2028 — *F.R.S.-FNRS Research Credit* (~12,000 €) for project **INSTAP** (*Infinite-State Synthesis: Theory and Practice*).

2024 — F.R.S.-FNRS *postdoctoral fellowship* (three years, to work with Emmanuel Filiot at Université libre de Bruxelles) — *declined in favor of the* Premier Assistant *position at UMONS*.

2019–2023 — F.R.S.-FNRS *research fellowship* (supervisors: Patricia Bouyer at Université Paris-Saclay and Mickael Randour at UMONS).

Honors

2025 *Outstanding Paper Award* at AAAI 2025 (awarded to 3 papers out of 3,000+ accepted papers).

2024 Selected to participate in the 11$^{\text{th}}$ Heidelberg Laureate Forum.

2024 Nominated for the *Best Paper Award* of CONCUR 2024.

2022 Prix Doctorants STIC 2022 – *accessit* (i.e., honorable mention), Labex DigiCosme.

2020 Nominated for the *Best Paper Award* of CONCUR 2020.

2019 Award from the Mathematics Department of UMONS for bachelor's and master's studies.

Publications

In theoretical computer science, authorship is traditionally listed alphabetically. Except for [ZVX+23], all my papers follow this convention.
Links to the articles (including extended versions on arXiv) are available on my website. You can also find my publications on DBLP, Google Scholar, and ORCID.

Peer-reviewed journal articles

[BCRV24] **Half-Positional Objectives Recognized by Deterministic Büchi Automata**. Patricia Bouyer, Antonio Casares, Mickael Randour, Pierre Vandenhove. *Logical Methods in Computer Science*, volume 20, issue 3, pages 19:1–19:42, 2024.

[BORV23] **Arena-Independent Finite-Memory Determinacy in Stochastic Games**. Patricia Bouyer, Youssouf Oualhadj, Mickael Randour, Pierre Vandenhove. *Logical Methods in Computer Science*, volume 19, issue 4, pages 18:1–18:51, 2023.

[BRV23] **Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs**. Patricia Bouyer, Mickael Randour, Pierre Vandenhove. *TheoretiCS*, volume 2, pages 1–48, 2023.

[ZVX+23] **Parallel and Memory-Efficient Distributed Edge Learning in B5G IoT Networks**. Jianxin Zhao, Pierre Vandenhove, Peng Xu, Hao Tao, Liang Wang, Chi Harold Liu, Jon Crowcroft. *IEEE Journal of Selected Topics in Signal Processing*, volume 17, issue 1, IEEE, pages 222–233, 2023.

[BBRRV22] **Decisiveness of Stochastic Systems and Its Application to Hybrid Models**. Patricia Bouyer, Thomas Brihaye, Mickael Randour, Cédric Rivière, Pierre Vandenhove. *Information and Computation*, volume 289, part B, Elsevier, pages 1–25, 2022.

[BLORV22] **Games Where You Can Play Optimally with Arena-Independent Finite Memory**. Patricia Bouyer, Stéphane Le Roux, Youssouf Oualhadj, Mickael Randour, Pierre Vandenhove. *Logical Methods in Computer Science*, volume 18, issue 1, pages 11:1–11:44, 2022.

Peer-reviewed conference proceedings and edited volumes

[BFMMV26] **LTL$_f$ Learning Meets Boolean Set Cover**. Gabriel Bathie, Nathanaël Fijalkow, Théo Matricon, Baptiste Mouillon, Pierre Vandenhove. *32nd International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2026)*, Lecture Notes in Computer Science 16505, Springer, pages 480–501, 2026.

[BBBFV25] **Decisiveness for Countable MDPs and Insights for NPLCSs and POMDPs**. Nathalie Bertrand, Patricia Bouyer, Thomas Brihaye, Paulin Fournier, Pierre Vandenhove. *Principles of Formal Quantitative Analysis (Essays Dedicated to Christel Baier on the Occasion of Her 60th Birthday)*, Lecture Notes in Computer Science 15760, Springer, pages 70–98, 2025.

[BFG+25] **Revelations: A Decidable Class of POMDPs with Omega-Regular Objectives**. Marius Belly, Nathanaël Fijalkow, Hugo Gimbert, Florian Horn, Guillermo A. Pérez, Pierre Vandenhove. *39th Annual AAAI Conference on Artificial Intelligence (AAAI 2025)*, pages 26454–26462, 2025. *Outstanding Paper Award (awarded to 3 papers out of 3,000+ accepted papers)*.

[BIPTV24] **The Power of Counting Steps in Quantitative Games**. Sougata Bose, Rasmus Ibsen-Jensen, David Purser, Patrick Totzke, Pierre Vandenhove. *35th International Conference on Concurrency Theory (CONCUR 2024)*, LIPIcs 311, Schloss Dagstuhl, pages 13:1–13:18, 2024. *Best Paper Award nomination*.

[BFRV23] **How to Play Optimally for Regular Objectives?** Patricia Bouyer, Nathanaël Fijalkow, Mickael Randour, Pierre Vandenhove. *50th EATCS International Colloquium on Automata, Languages and Programming (ICALP 2023)*, LIPIcs 261, Schloss Dagstuhl, pages 118:1–118:18, 2023.

[BCRV22] **Half-Positional Objectives Recognized by Deterministic Büchi Automata**. Patricia Bouyer, Antonio Casares, Mickael Randour, Pierre Vandenhove. *33rd International Conference on Concurrency Theory (CONCUR 2022)*, LIPIcs 243, Schloss Dagstuhl, pages 20:1–20:18, 2022.

[BRV22a] **Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs**. Patricia Bouyer, Mickael Randour, Pierre Vandenhove. *39th International Symposium on Theoretical Aspects of Computer Science (STACS 2022)*, LIPIcs 219, Schloss Dagstuhl, pages 16:1–16:16, 2022.

[BORV21] **Arena-Independent Finite-Memory Determinacy in Stochastic Games**. Patricia Bouyer, Youssouf Oualhadj, Mickael Randour, Pierre Vandenhove. *32nd International Conference on Concurrency Theory (CONCUR 2021)*, LIPIcs 203, Schloss Dagstuhl, pages 26:1–26:18, 2021.

[BBRRV20] **Decisiveness of Stochastic Systems and Its Application to Hybrid Models**. Patricia Bouyer, Thomas Brihaye, Mickael Randour, Cédric Rivière, Pierre Vandenhove. *11th International Symposium on Games, Automata, Logics, and Formal Verification (GandALF 2020)*, EPTCS 326, pages 149–165, 2020.

[BLORV20] **Games Where You Can Play Optimally with Arena-Independent Finite Memory**. Patricia Bouyer, Stéphane Le Roux, Youssouf Oualhadj, Mickael Randour, Pierre Vandenhove. *31st International Conference on Concurrency Theory (CONCUR 2020)*, LIPIcs 171, Schloss Dagstuhl, pages 24:1–24:22, 2020. *Best Paper Award nomination*.

Invited papers in international conferences

[BCRV23] **Half-Positional Objectives Recognized by Deterministic Büchi Automata (Extended Abstract)**. Patricia Bouyer, Antonio Casares, Mickael Randour, Pierre Vandenhove. *32nd International Joint Conference on Artificial Intelligence (IJCAI 2023), Sister Conferences Best Papers*, pages 6420–6425, 2023.

[BRV22b] **The True Colors of Memory: A Tour of Chromatic-Memory Strategies in Zero-Sum Games on Graphs**. Patricia Bouyer, Mickael Randour, Pierre Vandenhove. *42nd IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2022, invited talk by Patricia Bouyer)*, LIPIcs 250, Schloss Dagstuhl, pages 3:1–3:18, 2022.

Book chapter

[COV26] **Positionality and Memory**. Antonio Casares, Pierre Ohlmann, Pierre Vandenhove. Chapter 4 of the book **Games on Graphs: From Logic and Automata to Algorithms**, edited by Nathanaël Fijalkow. *Cambridge University Press*, pages 112–157, 2026.

Preprints

[FGKPV26] **Computing the Reachability Value of Posterior-Deterministic POMDPs**. Nathanaël Fijalkow, Arka Ghosh, Roman Kniazev, Guillermo A. Pérez, Pierre Vandenhove. 37 pages, 2026.

[COV24] **A positional $\mathbf{\Pi^0_3}$-complete objective**. Antonio Casares, Pierre Ohlmann, Pierre Vandenhove. 7 pages, 2024.

PhD Thesis

Title	**Strategy Complexity of Zero-Sum Games on Graphs** 268 pages, 2023.
Supervisors	Mickael Randour (UMONS) and Patricia Bouyer (LMF)
Committee	Christel Baier and Thomas Colcombet, reviewers; Véronique Bruyère, Laurent Doyen, and Benjamin Monmege, examiners; Patricia Bouyer and Mickael Randour, supervisors.
Defense	April 26, 2023

Talks

Slides are available on my website.

Invited seminars and talks

- Partially Observable MDPs: Revealing ones, and other decidable classes thereof (joint talk with Guillermo A. Pérez), *ULB Verif Seminar*, 12 Mar 2026, Brussels, Belgium.
- From probabilistic automata to POMDPs: Making Optimal Decisions Under Uncertainty, *Theoretical CS Seminar, Universiteit Antwerpen*, 11 Mar 2026, Antwerp, Belgium.
- The Decisiveness Property for Decidable Classes of Stochastic Systems, *Liverpool Verification Seminar*, 29 Feb 2024, Liverpool, UK.
- Characterizing Omega-Regularity through Strategy Complexity of Zero-Sum Games, *Young Scholar Day of the Belgian Mathematical Society*, 20 Dec 2023, Brussels, Belgium.
- Strategy Complexity of Zero-Sum Games on Graphs, *Links' Team Seminar*, 24 Nov 2023, Lille, France.
- Strategy Complexity of Zero-Sum Games on Graphs, *MoVe Seminar*, 9 Nov 2023, Marseille, France.
- Strategy Complexity of Zero-Sum Games on Graphs, *Liverpool Verification Seminar*, 3 July 2023, Liverpool, UK.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs, *STIC Doctoral Day on the Saclay plateau*, 20 June 2023, Gif-sur-Yvette, France.
- Strategy Complexity of Zero-Sum Games on Graphs, *Seminar of IST Austria*, 9 May 2023, Vienna, Austria.
- Memory Requirements of Omega-Regular Objectives: the Regular Case, *LaBRI/MTV Seminar*, 2 Mar 2023, Bordeaux, France.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite

Graphs, *IRIF Automata Seminar*, 28 Oct 2022, Paris, France.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs, *ULB Verif Seminar*, 14 Oct 2022, Brussels, Belgium.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs, *LaBRI/LX seminar*, 3 Mar 2022, Bordeaux, France.

Conference talks

- Revelations: A Decidable Class of POMDPs with Omega-Regular Objectives, *AAAI 2025: The 39th Annual AAAI Conference on Artificial Intelligence*, 2 Mar 2025, Philadelphia, PA, USA.
- Revelations: A Decidable Class of POMDPs with Omega-Regular Objectives, *Journées du GT Vérif*, 20 Nov 2024, Lille, France.
- Decidability of Omega-Regular Objectives for POMDPs with Revelations, *Dagstuhl Seminar 24231: Stochastic Games*, 4 June 2024, Dagstuhl, Germany.
- How to Play Optimally for Regular Objectives?, *Highlights 2023 of Logic, Games and Automata*, 25 July 2023, Kassel, Germany.
- How to Play Optimally for Regular Objectives?, *ICALP 2023: 50th EATCS International Colloquium on Automata, Languages and Programming*, 11 July 2023, Paderborn, Germany.
- Half-Positional Objectives Recognized by Deterministic Büchi Automata, *CONCUR 2022: The 33rd International Conference on Concurrency Theory*, 14 Sept 2022, Warsaw, Poland.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs, *LAMAS and SR 2022: Logical Aspects in Multi-Agent Systems and Strategic Reasoning*, 26 Aug 2022, Rennes, France.
- Half-Positional Objectives Recognized by Deterministic Büchi Automata, *Highlights 2022 of Logic, Games and Automata*, 29 June 2022, Paris, France.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs, *Current Trends in Graph and Stochastic Games (GAMENET Workshop)*, 6 Apr 2022, Maastricht, The Netherlands.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs, *STACS 2022: 39th International Symposium on Theoretical Aspects of Computer Science*, 17 Mar 2022, Virtual.
- Characterizing Omega-Regularity through Finite-Memory Determinacy of Games on Infinite Graphs, *Journées du GT Vérif*, 17 Nov 2021, ENS Paris-Saclay, Gif-sur-Yvette, France.
- Arena-Independent Finite-Memory Determinacy, *Highlights 2021 of Logic, Games and Automata*, 15 Sept 2021, Virtual.
- Arena-Independent Finite-Memory Determinacy in Stochastic Games, *CONCUR 2021: The 32nd International Conference on Concurrency Theory*, 26 Aug 2021, Virtual.
- Arena-Independent Finite-Memory Strategies, *GT ALGA – Journées annuelles 2021*, 17 June 2021, Virtual.
- Decisiveness of Stochastic Systems and Its Application to Hybrid Models, *11th International Symposium on Games, Automata, Logics, and Formal Verification (GandALF 2020)*, 22 Sept 2020, Virtual.
- Games Where You Can Play Optimally with Arena-Independent Finite Memory, *CONCUR 2020: The 31st International Conference on Concurrency Theory*, 2 Sept 2020, Virtual.
- Games Where You Can Play Optimally with Arena-Independent Finite Memory, *MOVEP 2020: 14th Summer School on Modelling and Verification of Parallel Processes*, 22 June 2020, Virtual.
- Reachability in Stochastic Hybrid Systems, *Highlights 2019 of Logic, Games and Automata*, 19 Sept 2019, Warsaw, Poland.

- Reachability in Stochastic Hybrid Systems, *13th International Conference on Reachability Problems (RP 2019)*, 12 Sept 2019, Brussels, Belgium.

Local talks

- Martingale Theory for the Average MDP Enjoyer, *UMONS Formal Methods Reading Group*, 1 Dec 2025, Mons, Belgium.
- Revelations: A Decidable Class of POMDPs with Omega-Regular Objectives, *INFORTECH Day 2025*, 23 May 2025, Mons, Belgium.
- Revelations: A Decidable Class of POMDPs with Omega-Regular Objectives, *UMONS Formal Methods Reading Group*, 7 Oct 2024, Mons, Belgium.
- How to Play Optimally for Regular Objectives?, *GT Informel CDS/MCS of the Laboratoire Méthodes Formelles*, 17 Feb 2023, Gif-sur-Yvette, France.
- Memory Requirements of Omega-Regular Objectives: the Regular Case, *UMONS Formal Methods Reading Group*, 16 Dec 2022, Mons, Belgium.
- Existence of memoryless optimal strategies through universal graphs [based on a LICS'22 paper], *UMONS Formal Methods Reading Group*, 2 June 2022, Mons, Belgium.
- Characterizing Omega-Regularity through Strategy Complexity of Games on Infinite Graphs [Ongoing Work], *UMONS Formal Methods Reading Group*, 23 Sept 2021, Mons, Belgium.
- Arena-Independent Finite-Memory Strategies, *GT Model-Checking and Synthesis, LMF, Université Paris-Saclay*, 23 Apr 2021, Virtual.
- Understanding Finite-Memory Determinacy, *LMF Research Days*, 9 Dec 2020, Virtual.
- Reachability in Infinite Markov Chains, *Mardi des Chercheurs*, 5 Mar 2019, Université de Mons, Mons, Belgium.

Thesis talks and defenses

- Strategy Complexity of Zero-Sum Games on Graphs, *public thesis defense*, 26 Apr 2023, Mons, Belgium.
- Strategy Complexity of Zero-Sum Games on Graphs, *private thesis defense*, 20 Apr 2023, Mons, Belgium.
- Strategy Complexity of Zero-Sum Games on Graphs (Thesis seminar), *LMF Seminar*, 14 Mar 2023, Gif-sur-Yvette, France.

Popular Science and Outreach

Articles in General-Interest Media

- **Making optimal decisions without having all the cards in hand**. Nathanaël Fijalkow, Hugo Gimbert, Florian Horn, Guillermo A. Pérez, Pierre Vandenhove. *AIHub*, 24 June 2025. (article).
- **Prendre des décisions optimales sans avoir toutes les cartes en main**. Pierre Vandenhove. *Le Monde: Blog Binaire*, 16 May 2025. Two-part article. (part 1, part 2).

Talks

- Erreur détectée : Comment les ordinateurs réparent toutes nos erreurs (ou presque !), *Popular science talk for secondary school students at Collège Notre-Dame de Tournai*, 13 Oct 2025, Tournai, Belgium.
- Strategy Complexity: How Much Does It Take to Win?, *Brussels Summer School of Mathematics*, 26 Aug 2025, Brussels, Belgium.
- Erreur détectée : Comment les ordinateurs réparent toutes nos erreurs (ou presque !), *Journées Math–Sciences 2025*, 27–28 Mar 2025, Mons, Belgium.

– Jeux pour l'informatique et complexité des stratégies, *Séminaire Jeunes de l'UMONS*, 21 Apr 2022, Mons, Belgium.
– À vous de jouer !, *Journées Math–Sciences*, 24 Mar 2022, Mons, Belgium.

Teaching

Feb 2025–Present	**Introduction to Machine Learning and Data Science**, *Lecturer*, Université de Mons Lectures and practical sessions (about 60 h/year).
Sept 2025–Present	**Data Structures I**, *Lecturer*, Université de Mons Lectures (about 24 h/year).
Sept 2024–Present	**Operating Systems**, *Lecturer*, Université de Mons Lectures (taught twice annually, about 54 h/year).
Jan 2024–June 2024	**Machine Learning and Deep Learning**, *Teaching Assistant*, Université de Bordeaux Lectures and practical sessions (about 18 h).
Jan 2024–June 2024	**Computer Science**, *Teaching Assistant*, Lycée Montaigne (CPGE) Algorithms and programming exercises (about 24 h).
Sept 2019–Aug 2023	**Formal Methods for System Design**, *Teaching Assistant*, Université de Mons Practical sessions (about 50 h/year).

Student supervision

Interns

– Co-supervision (with Thomas Brihaye) of David Chandellier on the safety value in POMDPs (June to July 2026).
– Co-supervision (with Thomas Brihaye) of Xavier Goubault–Larrecq on the safety value in POMDPs (June to July 2026).
– Co-supervision (with Sougata Bose and Mickael Randour) of Léonard Bœuf on the strategy complexity of games on infinite arenas (February to July 2026).
– Supervision of Matéo Torrents on memory requirements of stochastic games (June to July 2024).
– Co-supervision (with Gabriel Bathie, Nathanaël Fijalkow, and Théo Matricon) of Baptiste Mouillon on LTL learning (March to July 2024).
– Co-supervision (with Nathanaël Fijalkow, Guillaume Lagarde, and Théo Matricon) of Sylvain Brisset on programmatic reinforcement learning (February to April 2024).
– Co-supervision (with Nathanaël Fijalkow and Théo Matricon) of Gianni Padula on incremental reactive synthesis (November 2023 to February 2024).
– Co-supervision (with Mickael Randour) of Jean Abou Samra on automata minimization (June to August 2023).
– Supervision of Maximilien Vanhaverbeke on succinctness of good-for-games automata (August 2022).
– Supervision of Luca Lani on strategy complexity of zero-sum games (August 2021).

Supervision of master's theses

Clément Allard (2026, UMONS), Nicolas Melaerts (2026, UMONS), Xavier Goubault–Larrecq (2026, ENS Paris), Amar Hamouma (2025, UMONS), Jordan Demaret (2025, UMONS), Matéo Torrents (2024, ENS Paris).

Jury member for master's theses

Nathan Amorison (2025), Stéphane Délire (2025), Antoine Guns (2025), Valentin Dusollier (2023), James C. A. Main (2021). All at UMONS.

Attended events

2026

IJCAI-ECAI 2026, Bremen, Germany.

2025

Brussels Summer School of Mathematics, Brussels, Belgium | *INFORTECH Day 2025*, Mons, Belgium | *AAAI 2025: The 39th Annual AAAI Conference on Artificial Intelligence*, Philadelphia, PA, USA.

2024

Journées du GT Vérif, Lille, France | *Dagstuhl Seminar 24231: Stochastic Games*, Dagstuhl, Germany | *Journées du GT DAAL*, Rennes, France.

2023

Belgian Mathematical Society Young Scholar Day, Brussels, Belgium | *Highlights 2023 of Logic, Games and Automata*, Kassel, Germany | *ICALP 2023: 50th EATCS International Colloquium on Automata, Languages and Programming*, Paderborn, Germany | *STIC Doctoral Day on the Saclay plateau*, Gif-sur-Yvette, France.

2022

CONCUR 2022: The 33rd International Conference on Concurrency Theory, Warsaw, Poland | *LAMAS and SR 2022: Logical Aspects in Multi-Agent Systems and Strategic Reasoning*, Rennes, France | *Highlights 2022 of Logic, Games and Automata*, Paris, France | *Current Trends in Graph and Stochastic Games (GAMENET Workshop)*, Maastricht, The Netherlands | *39th International Symposium on Theoretical Aspects of Computer Science (STACS 2022)*, Virtual.

2021

Journées du GT Vérif, ENS Paris-Saclay, Gif-sur-Yvette, France | *Highlights 2021 of Logic, Games and Automata*, Virtual | *CONCUR 2021: The 32nd International Conference on Concurrency Theory*, Virtual | *Reinforcement Learning – From Theory to Practice Summer School*, Alan Turing Institute | *GT ALGA – Journées annuelles 2021*, Virtual.

2020

Spotlight on Games, Virtual | *GandALF 2020: 11th International Symposium on Games, Automata, Logics, and Formal Verification*, Virtual | *Highlights 2020 of Logic, Games and Automata*, Virtual | *CONCUR 2020: The 31st International Conference on Concurrency Theory*, Virtual | *MOVEP 2020: 14th Summer School on Modelling and Verification of Parallel Processes*, Virtual.

2019

Highlights 2019 of Logic, Games and Automata, Warsaw, Poland | *13th International Conference on Reachability Problems (RP 2019)*, Brussels, Belgium | *Theory and Algorithms in Graph and Stochastic Games*, Mons, Belgium | *Mardi des Chercheurs*, Mons, Belgium.

2018

FoPSS Logic and Learning School, Oxford, UK | *MOVEP 2018: 13th Summer School on Modelling and Verification of Parallel Processes*, Cachan, France | *International Conference on Functional Programming (ICFP 2018)*, St. Louis, Missouri, United States.

2017

Computers in Scientific Discoveries 8, Mons, Belgium.

Internships

Aug–Nov 2018 **Student intern**, *OCaml Labs, University of Cambridge*, UK
Implementation of the Mask R-CNN architecture for image segmentation and classification using OCaml's numerical library Owl. Contributions to the *computation graph* module.

Jul 2017 **Research intern**, *Algorithms Lab, UMONS*, Belgium
Received a grant for a research internship in graph theory about coloration problems.

Miscellaneous

Program committees: Program committee member for CONCUR 2026, AAAI 2026.

Reviewing: Subreviewer for Logical Methods in Computer Science (x3), Dynamic Games and Applications, GandALF, LICS (x4), ICALP (x4), STACS, CONCUR (x5), FoSSaCS (x2), FSTTCS, CSL, MFCS, ATVA, FORMATS, SETTA, EUMAS, VMCAI, LATIN, MathSciNet (x4).

Academic service:
- Member of the Council of the Faculty of Science at UMONS (2020–2023, and since 2024).
- Co-organizer of the *MTV seminar* at LaBRI (academic year 2023–2024).
- Member of the organizing committee of Highlights 2024 of Logic, Games and Automata at LaBRI, Université de Bordeaux; responsible for the Highlights Extended Stay Support Scheme.

Programming contests:
- UMONS representative for the BeOI (*Belgian Olympiad in Informatics*) for secondary schools since 2025. Organizer of a two-day training session in March 2025. Organizer of an annual local qualification center.
- Coach of the ICPC teams at UMONS, and organizer of the *BAPC Preliminaries* at UMONS in 2019, 2020, 2021, 2022, and 2024.
- Participation in ICPC (programming contests) as a student from 2016 to 2018. Our team CPUMONS won a bronze medal (rank 12/114) during NWERC 2016, 3rd place during BAPC 2016 and BAPC 2018, 5th place during BAPC 2017.